EDAM Bioimaging

Enabling findability and provenance of instruments, techniques, data, and analysis tools

Adriaan Ludl[1], Laure Plantard[2], Joakim Lindblad[3], Martin Jones[4], Melanie Föll[5], Núria Queralt Rosinach[6], Johan Gustafsson[7], Angelika Svetlove[8], Cynthia Arend[9], Leonid Kostrykin[9], Alexandra Zakieva[10], Helge Hecht[11], Anna Seybold[12], Harald Hausen[1], Moritz Kirschmann[13], Anatole Chessel[14], Leandro Scholz[15*], Fabienne Rössler[13*], Laura Nicolás Sáenz[16*], Estibaliz Gómez de Mariscal[16*], John Bogovic[17], Alexandre Dufour[18*], Rohola Hosseini[19], Xavier Heiligenstein[20*], Dominic Waithe[21*], Marie-Charlotte Domart[4], Matthia Winter-Karreman[22], Raf Van de Plas[23], Robert Haase[24*], David Hörl[25], Lassi Paavolainen[26], Paula Sampaio[27], Dean Karaica[28*], Ivana Vrhovac Madunić[28], Arrate Muñoz-Barrutia[16], Daniel Sage[29], Sebastian Munck[30], Ofra Golani[31], Federico Bianchini[32], Josh Moore[33], Florian Levet[34], Nataša Sladoje[3], Chong Zhang[35], Magnus Palmblad[6], Kota Miura[9], Beatriz Serrano-Solano[36], Teresa Zulueta-Coarasa[37], Julien Colombelli[35], Caterina Strambio De Castillia[38], Perrine Paul-Gilloteaux[39], and Matúš Kalaš[1]✉.* ***EDAM Bioimaging is welcoming new contributors!***

*[1]University of **Bergen, Norway**; [2]Friedrich Miescher Institute for Biomedical Research, **Basel, Switzerland**; [3]Centre for Image Analysis, **Uppsala** University, **Sweden**; [4]EM platform, Francis Crick Institute, **London, UK**; [5]University Medical Center **Freiburg, Germany**; [6]**Leiden** University Medical Center, **Netherlands**; [7]**Australian BioCommons**, & University of **Melbourne, Australia**; [8]EMBL **Hamburg, Germany**; [9]University of **Heidelberg, Germany**; [10]EMBL **Heidelberg, Germany**; [11]Masaryk University, **Brno, Czechia**; [12]University of **Innsbruck, Austria**; [13]University of **Zürich, Switzerland**; [14]École Polytechnique, **Palaiseau, France**; [15]Federal University of Paraná, **Curitiba, Brazil**; [16]Universidad Carlos III de **Madrid, Spain**; [17]Janelia Research Campus, **Ashburn, VA, USA**; [18]Institut Pasteur, **Paris, France**; [19]**Leiden** University, **Netherlands**; [20]Institut Curie, **Paris, France**; [21]WIMM, University of **Oxford, UK**; [22]German Cancer Research Center (DKFZ), **Heidelberg, Germany**; [23]**Delft** University of Technology, **Netherlands**; [24]Max Planck Institute of Molecular Cell Biology and Genetics, **Dresden, Germany**; [25]LMU **Munich, Germany**; [26]FIMM, University of **Helsinki, Finland**; [27]University of **Porto, Portugal**; [28]Institute for Medical Research and Occupational Health, **Zagreb, Croatia**; [29]École Polytechnique Fédérale de **Lausanne, Switzerland**; [30]VIB, and KU **Leuven, Belgium**; [31]Weizmann Institute, **Rehovot, Israel**; [32]University of **Oslo, Norway**; [33]**OME**, and **German BioImaging, Germany**; [34]University of **Bordeaux, France**; [35]Institute for Research in Biomedicine, **Barcelona, Spain**; [36]Euro-BioImaging BioHub, EMBL **Heidelberg, Germany**; [37]EMBL-EBI, **Hixton, UK**; [38]University of Massachusetts Chan Medical School, **Worcester, MA, USA**; [39]University of **Nantes, France**.*
**at the time of contribution*

✉ drmatuskalas@duck.com

/edamontology/edam-bioimaging

2026 CC BY

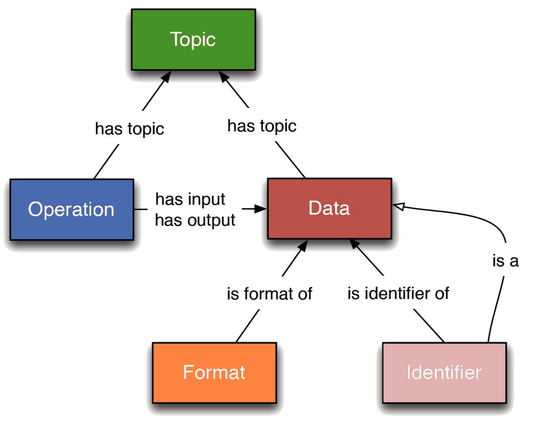
What? EDAM Bioimaging is an extension of the EDAM ontology, dedicated to (bio)imaging, image analysis, image processing, and bioimage informatics.

Why? EDAM Bioimaging fosters open science by enabling interoperable descriptions of techniques, instruments, software, data, publications, training, *etc.*

How? EDAM Bioimaging is developed in an open community spirit, in a welcoming collaboration between imaging experts, bioscientists, and engineers.

How can I contribute? EDAM needs your expertise! Please help by posting comments with suggestions or needs for clarification (here), or by creating GitHub issues, or by organising in a collaborative event. Please see https://github.com/edamontology/edam-bioimaging#contributing.

STRUCTURE OF EDAM



EXAMPLE CONCEPTS

rdfs:label	Correlative light and electron microscopy
rdfs:seeAlso	https://en.wikipedia.org/wiki/Correlative_light-electron_microscopy
hasExactSynonym	Correlative light-electron microscopy
hasNarrowSynonym	Integrated light-electron microscopy
hasNarrowSynonym	Integrated light and electron microscopy (ILEM)
hasNarrowSynonym	Volume correlative light and electron microscopy (Volume CLEM)
Abbreviation	CLEM
exactMatch	https://www.wikidata.org/wiki/Q28457249
hasDefinition	Correlative light and electron microscopy is the combination of lig (typically fluorescence microscopy) and electron microscopy of th
inSubset	topics
inSubset	bio
inSubset	imaging
Enter property	Enter value

Parents
- Light microscopy
- Electron microscopy
- Multimodal imaging

Enter a class name

Relationships
- can use — Array tomography

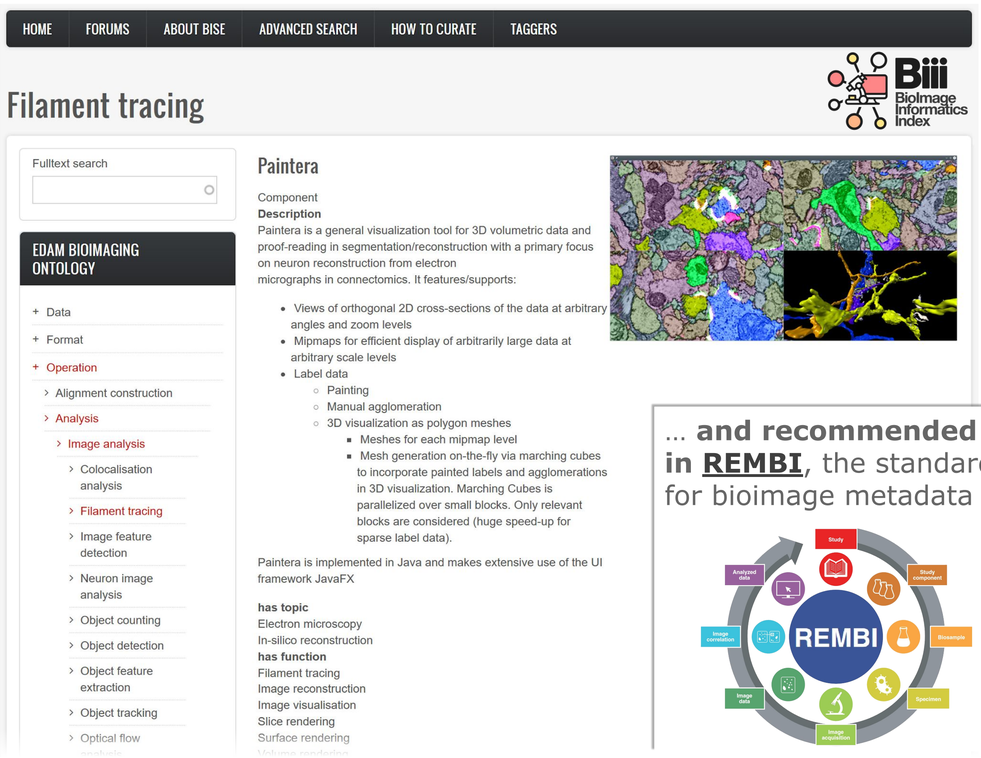
rdfs:label	Filament tracing
rdfs:comment	Filament tracing is an image analysis operation which consumes an image of a filamentous structure (it may be a tree-like structure, a filament network or a agglomeration of single 'stick-like' filaments), and outputs data that represent the filament(s), most commonly a skeleton representation of the filaments and their diameters or surfaces.
hasExactSynonym	Tubular structure extraction
hasRelatedSynonym	Curvilinear structure detection
hasRelatedSynonym	Curvilinear structure reconstruction
hasNarrowSynonym	Biofilament tracing
inSubset	bio
inSubset	imaging
inSubset	operations
Related term	Neuron reconstruction
Enter property	Enter value

Parents
- Image segmentation

Enter a class name

Relationships
- related — Neuron image analysis

EDAM Bioimaging is used *e.g.* in Biii.eu, the registry of bioimage analysis tools, workflows, and training materials …



… **and recommended in REMBI**, the standard for bioimage metadata

HIERARCHIES OF CONCEPTS IN EDAM Bioimaging

- Topic (1)
  - Artificial intelligence (17)
  - Biosciences
  - Chemistry
  - Computational linguistics
  - Data acquisition
    - Cytometry
    - Diffractometry and scattering
    - Imaging (5)
      - Bioimaging
      - Bioluminescence imaging
      - Echography
      - Electrocardiography
      - Electroencephalography
      - Fixed-sample imaging
      - High-content analysis
      - Imaging cytometry
      - Live sample imaging (1)
      - MRI (2)
      - Mass spectrometry imaging (4)
      - Microscopy
        - Adaptive microscopy (3)
        - Electron microscopy (1)
          - CLEM (3)
          - CLXEM
          - CXEM
          - Cryo-EM (1)
          - Cryogenic electron microscopy
          - EDS
          - EELS
          - Electron diffractometry and scattering
          - In situ electron microscopy
          - Sample thinning
          - Scanning electron microscopy (2)
          - Transmission electron microscopy
          - Volume electron microscopy
        - Light microscopy (1)
          - Array detection microscopy
          - CLEM (3)
          - CLXEM
          - CLXM
          - Cryo-LM
          - Fluorescence microscopy
            - Confocal microscopy (1)
            - FCS
            - FISH (1)
            - FLIM
            - FRAP
            - FRET (1)
            - Image correlation spectroscopy (1)
            - Light sheet fluorescence microscopy (4)
            - Multi-photon microscopy (2)
            - Ratiometric imaging (1)
            - Super-resolution microscopy (3)
            - TIRF
            - Widefield microscopy
          - Imaging flow cytometry (2)
          - Interferometric microscopy
          - Label-free microscopy (2)
          - Light field microscopy
          - Non-linear microscopy
          - Photomanipulation
          - Point detection microscopy
          - Point-scanning microscopy (1)
          - Raman microscopy
          - Stereoscopy
          - Transmission light microscopy (1)
        - Photonic force microscopy
        - Scanning probe microscopy
        - Traction force microscopy
        - X-ray microscopy
      - Multimodal imaging (4)
      - Spectroscopy (1)
      - Tissue imaging (3)
      - Tomography
        - Ultrasonography (1)
        - X-ray imaging (3)
    - Single-cell data acquisition
    - Spectrometry (1)
  - Data management
  - High-throughput screening
  - Histology
  - In-silico reconstruction (1)
  - Instrument maintenance
  - Open science
  - Sample material separation (1)
  - Sample preparation (2)
  - Scientific visualisation (1)
  - Statistics and probability (2)
- Operation (2)
  - Alignment construction
    - Image registration (6)
  - Annotation
  - Calculation
  - Classification
    - Image classification (3)
    - Object classification
    - Pixel classification
  - Clustering
  - Comparison
  - Correlation
  - Data analysis
    - Frequency-domain analysis
    - Image analysis (4)
      - Cell migration analysis
      - Colocalisation analysis
      - Image classification (3)
      - Image feature detection
      - Image segmentation (4)
        - Cell segmentation (1)
        - Connected-component analysis (3)
        - Filament tracing (1)
        - Image thresholding
        - Interactive segmentation
        - Ion image segmentation
        - Model-based segmentation (3)
        - Overlap analysis
        - Pixel classification
        - Region growing
        - Superpixel extraction
      - Neurite measurement
      - Object classification
      - Object counting
      - Object detection
      - Object feature extraction
        - Homogeneity extraction
        - Overlap analysis
        - Shape features extraction
        - Stereology
        - Texture extraction
        - Velocity extraction
      - Object tracking
        - Cell lineage extraction
        - Collective object tracking
        - Isolated object tracking
        - Trajectory analysis
        - Wound-healing analysis (1)
      - Optical flow analysis (2)
    - Spatial analysis
    - Spectral analysis
    - Structure analysis
    - Temporal analysis
  - Data generation
  - Data handling
  - Data management planning
  - Design
  - Image processing (4)
    - Frequency-domain transformation
    - Geometric transformation (4)
    - Image convolution
    - Image correction (2)
      - Chromatic aberration correction
      - Detector bias correction
      - Detector gain correction
      - Drift correction
      - Geometric distortion correction (1)
      - Illumination correction (2)
      - Spectral unmixing
    - Image crop
    - Image enhancement (9)
    - Image fusion
    - Image reconstruction
    - Image registration (6)
    - Image segmentation (4)
    - Image-to-image translation (3)
    - Ion image normalisation
    - Morphological operation (1)
  - Indexing
  - Mapping
  - Modelling and simulation
  - Optimisation and refinement
  - Optimisation or refinement
  - Pattern recognition
  - Quantification
  - Regression (2)
  - Service management
  - Validation (1)
  - Visualisation (2)
- Data (3)
  - Data cube
  - Data management plan
  - Experimental measurement
  - Identifier
  - Image annotation data
    - Class labels (1)
    - Derived annotations
    - Image annotation graph
    - Object counts
    - Object trajectory
    - Point annotations
    - ROI
      - Bounding box
      - Geometrical annotations
      - Segmentation mask
  - Image characteristic
  - Image data (2)
    - 1D image?
    - 2D image
    - 3D image
    - Cell migration track image
    - Fate map
    - Image time series
    - Ion image
    - Multi-channel image (1)
    - Raw image
    - nD image?
  - Mass spectrometry imaging data
  - Metadata
  - Object characteristics
  - Region property
  - Report
  - Spectrum
  - Structure
  - Time-lapse data
- Format (4)
  - Binary format
    - AVI
    - DICOM format
    - GIF
    - HDF5
    - JPG
    - LSM
    - MPEG-4
    - Nikon NIS-Elements ND2
    - PNG
    - TIFF
      - OME-TIFF
    - WMV
    - Zarr
      - OME-Zarr
    - Zeiss CZI
    - hdr
    - ibd
    - ims
    - nii
    - pbm
    - pcd
    - pcx
    - pgm
    - ppm
    - psd
    - raw
    - rgb
    - scn
    - sm2
    - sm3
    - stk
    - vsi
    - xbm
    - xpm
  - Format (by type of data)
  - ISA-JSON
  - Textual format
  - XML
    - OME-XML
    - SVG
    - imzML metadata file
    - nmrML
  - imzML

Browse interactively in WebProtégé (free registration)

Developed with support from:

In collaboration and coordination with members from: